

Geothermal Power Generation Business Plan for Rwanda

Executive Summary

This business plan outlines a comprehensive strategy for developing geothermal power generation capacity in Rwanda, leveraging the country's significant geothermal resources to meet growing electricity demand, reduce dependence on fossil fuels, and support national development goals. The plan focuses on a phased approach that begins with resource validation and pilot projects, progressing to commercial-scale power plants that will contribute to Rwanda's target of increasing installed capacity from 216 MW to 512 MW by 2024 and achieving 100% electricity access.

Key Opportunity

Rwanda possesses conservative geothermal estimates of up to 100 MW, with some sources suggesting potential exceeding 700 MW. The Northwestern Volcanic Area (Gisenyi, Karisimbi, Kinigi) and Southern Rift-Fault Area (Bugarama) represent the most promising prospect zones for development.

Business Model

The venture will operate as an Independent Power Producer (IPP) developing, owning, and operating geothermal power plants under long-term Power Purchase Agreements (PPAs) with Rwanda Energy Group (REG) or direct sales to industrial customers. Revenue will be primarily generated through electricity sales at regulated tariffs, supplemented by potential carbon credit revenues and ancillary services. The business model also incorporates micro-turbine technology (1 kW to 50 kW) from Turtle Turbines for smaller-scale distributed generation applications, particularly suitable for remote communities, industrial waste heat recovery, and hybrid renewable systems.

Financial Highlights

- **Initial Investment:** \$16.25-\$32.5 million for a 5-10 MW binary cycle plant
- **Annual Revenue:** \$5.95-\$11.9 million (for 5-10 MW plant)
- **Simple Payback Period:** 2.7 years (after exploration risk mitigation)
- **25-Year NPV:** \$55-\$150 million (8% discount rate)
- **Job Creation:** 3-5 direct operations jobs per MW, plus indirect employment

Implementation Timeline

- **Phase 1 (Q4 2026-Q2 2027):** Foundation building, resource validation, partnerships

- **Phase 2 (Q3 2027-Q4 2028):** Pilot implementation, exploratory drilling
- **Phase 3 (2029 and beyond):** Scale and replicate, commercial plant commissioning

1. Company Overview

1.1 Vision

To become Rwanda's leading geothermal power producer, providing clean, reliable, and affordable baseload electricity to support the nation's industrialization and sustainable development goals.

1.2 Mission

To develop Rwanda's geothermal resources through innovative technology, strategic partnerships, and environmentally responsible practices, delivering superior returns to stakeholders while advancing national energy security.

1.3 Core Values

- Safety and environmental stewardship
- Technical excellence and innovation
- Partnership and collaboration
- Transparency and integrity
- Long-term value creation

1.4 Legal Structure

The venture will be established as a Rwandan limited liability company, with potential for international investment partners and potential listing on the Rwanda Stock Exchange as it matures.

2. Market Analysis

2.1 Rwanda's Energy Context

Current Situation

- **Installed Capacity:** ~216 MW (as of MININFRA data)
- **Target Capacity by 2024:** 512 MW (requiring ~296 MW of new capacity)
- **Electricity Access:** 51% of households (37% grid, 14% off-grid)
- **Access Goal:** 100% by 2024 (52% on-grid, 48% off-grid)
- **Current Reliance:** Significant dependence on diesel generators and imported electricity

Demand Drivers

- **Economic Growth:** Rwanda's Vision 2020 and National Strategy for Transformation (NST1) targeting middle-income status

- **Industrialization:** Made in Rwanda initiative and Kigali Innovation City
- **Urbanization:** Rapid growth of Kigali and secondary cities
- **Regional Integration:** Participation in East African Power Pool (EAPP)
- **Climate Commitments:** 38% emissions reduction pledge by 2030 under Paris Agreement

2.2 Geothermal Resource Assessment

Potential Estimates

- **Conservative:** Up to 100 MW (Rwanda Energy Group, African Geothermal Association)
- **Optimistic:** Potential exceeding 700 MW (Rwanda Dispatch)

Prospect Zones

1. **Northwestern Volcanic Area**
 - Gisenyi: Hot springs at 70-75°C
 - Karisimbi: Previous drilling (2006) - no viable system found
 - Kinigi: Prospect zone requiring detailed investigation
2. **Southern Rift-Fault Area**
 - Bugarama: Gradient-well work ongoing, primary current focus
3. **Additional Areas**
 - Near Lake Kivu
 - Volcanoes National Park periphery
 - Rubavu Graben
 - Mashyuza

Resource Characteristics

- **Temperature Range:** 70-75°C+ at surface manifestations (suitable for binary cycle)
- **Depth to Resource:** To be confirmed through exploration drilling
- **Fluid Chemistry:** Requires assessment for scaling/corrosion potential
- **Permeability:** Fracture-controlled systems expected in volcanic terrain

2.3 Competitive Landscape

Current Energy Mix

- Hydropower: ~50% of generation (seasonally variable)
- Diesel: ~30% (high operating costs)
- Methane (Lake Kivu): ~10%
- Solar: Growing but still small percentage
- Imports: Variable from neighboring countries

Competitive Advantages of Geothermal

- **Baseload Capacity:** 90-95% capacity factor vs. 20-30% for solar, 40-60% for hydro
- **Predictable Output:** Not weather-dependent
- **Low Operating Costs:** Minimal fuel costs after capital investment
- **Long Lifespan:** 25-30+ years with proper maintenance
- **Small Footprint:** Less land-intensive than solar or hydro per MW
- **Local Resource:** Reduces foreign exchange exposure for fuel imports

Market Positioning The venture will position itself as a provider of clean, reliable baseload power that complements Rwanda's existing hydro resources and reduces dependence on expensive diesel generation. Long-term PPAs will provide revenue certainty while supporting national grid stability.

3. Technology Description

3.1 Selected Technology: Binary Cycle Organic Rankine Cycle (ORC)

Given the resource temperature range of 70-75°C+ identified at surface manifestations, binary cycle technology is the most appropriate for Rwanda's geothermal resources.

How Binary Cycle Works

1. Hot geothermal fluid is brought to surface through production wells
2. Heat is transferred to a secondary working fluid (typically isobutane or similar organic compound) with a lower boiling point than water
3. The secondary fluid vaporizes, driving a turbine connected to an electrical generator
4. The working fluid is condensed and recycled in a closed loop
5. The cooled geothermal fluid is reinjected into the reservoir

Advantages for Rwanda

- **Temperature Suitability:** Efficient operation at 70-150°C range
- **Environmental Protection:** Zero emissions, closed-loop system
- **Proven Technology:** Commercial success in similar temperature resources globally
- **Modular Design:** Facilitates phased implementation
- **Grid Compatibility:** Standard electrical output compatible with REG infrastructure

3.3 Micro-Turbine Technology Options

For smaller-scale applications and distributed generation, the venture will also evaluate micro-turbine technology from providers such as Turtle Turbines (1 kW to 50 kW range). These systems can be particularly valuable for: - Remote community power systems - Industrial waste heat recovery applications - Hybrid

renewable energy systems (combined with solar or biomass) - Pilot projects to demonstrate technology viability before scaling to larger binary cycle plants - Applications requiring rapid deployment and modular scalability

Micro-turbines offer advantages including lower initial capital costs, faster deployment timelines, and suitability for applications where full-scale binary cycle plants may not be economically viable due to resource limitations or grid access constraints.

3.2 Plant Components

Surface Facilities

- Production and injection wells
- Heat exchangers (primary and secondary)
- Organic Rankine Cycle power block (turbine, generator, condenser)
- Cooling system (air-cooled condensers appropriate for Rwanda's climate)
- Electrical equipment (transformers, switchgear, controls)
- Reinjection system
- Ancillary facilities (control building, storage, workshops)

Subsurface Components

- Production wells targeting fracture zones
- Injection wells for reservoir pressure maintenance
- Downhole pumps (if natural insufficient)
- Well casings and cementing for zonal isolation
- Monitoring equipment (temperature, pressure, flow)

3.3 Technology Partners and Best Practices

The venture will leverage Kenya's extensive geothermal experience, adopting proven best practices:

Drilling Best Practices (from Kenya)

- **Phased Approach:** Slim holes (6-8.5") for exploration, followed by production wells (9.5-12.25") after confirmation
- **Directional Drilling:** Target specific fracture zones, increasing productivity by 30-50%
- **Managed Pressure Drilling (MPD):** Handle high-pressure zones safely
- **Lost Circulation Management:** Specialized materials for volcanic formations
- **Real-Time Monitoring:** Downhole sensors for optimization and early issue detection

Power Generation Best Practices

- **Steam Field Management:** Pressure/temperature tracking, tracer testing, production profiling
- **High-Efficiency Separators:** >99.5% steam purity protection for turbines
- **NCG Handling:** Steam ejectors or compressors to maintain turbine efficiency
- **Modular Design:** Standardized 150-350 MW units for phased expansion
- **Brines Management:** Reinjection strategies to maintain reservoir pressure

4. Marketing and Sales Strategy

4.1 Target Customers

Primary: Rwanda Energy Group (REG) through Power Purchase Agreements (PPAs) Secondary: Large industrial customers seeking reliable, clean power (mines, manufacturing) Tertiary: Off-grid or mini-grid applications for remote communities

4.2 Pricing Strategy

- **PPA Tariffs:** Negotiated with REG based on avoided cost of diesel generation
- **Escalation Clauses:** Tied to inflation indices (similar to Kenyan PPAs)
- **Take-or-Pay Provisions:** Ensure revenue certainty
- **Ancillary Services:** Potential revenue from frequency regulation, spinning reserve

4.3 Sales Process

1. **Resource Validation:** Demonstrate measured flow and temperature
2. **Technical Studies:** Grid impact, interconnection studies
3. **PPA Negotiation:** Standardized terms based on REG/RURA frameworks
4. **Financing Securement:** Commitments from lenders and investors
5. **EPC Contracting:** Fixed-price, turnkey construction
6. **Commissioning and Commercial Operation:** Performance testing and handover

4.4 Differentiation Factors

- **Reliability:** Baseload power with >90% availability
- **Clean Energy:** Zero operational emissions supporting climate goals
- **Local Development:** Job creation, technology transfer, economic stimulation
- **Price Stability:** Not subject to international fuel price volatility
- **Long-Term Partnership:** 25-30 year asset life supporting sustained collaboration

5. Operations Plan

5.1 Plant Operations

- **Staffing:** 3-5 direct operations personnel per MW (shift-based)
- **Maintenance:** Preventive and predictive maintenance programs
- **Monitoring:** SCADA systems for real-time performance tracking
- **Environmental Compliance:** Monitoring of reinjection, surface facilities, emissions
- **Safety Programs:** Comprehensive HSE management systems

5.2 Resource Management

- **Reservoir Monitoring:** Pressure, temperature, flow rates, chemical composition
- **Reinjection Strategy:** Maintain reservoir pressure and sustainability
- **Make-up Well Planning:** Reserve for reservoir decline compensation
- **Production Optimization:** Balance extraction with reservoir capacity

5.3 Maintenance Philosophy

- **Preventive Maintenance:** Regular scheduled maintenance based on OEM recommendations
- **Predictive Maintenance:** Condition monitoring, vibration analysis, oil analysis
- **Corrective Maintenance:** Rapid response to failures with spare parts inventory
- **Turnaround Planning:** Scheduled major maintenance every 3-5 years

5.4 Quality and Standards

- **International Standards:** Adherence to IEC, ISO, and relevant geothermal industry standards
- **Local Compliance:** RURA, REMA, and other Rwandan regulatory requirements
- **Best Practices:** Continuous improvement through benchmarking with global leaders
- **Certifications:** Pursue relevant quality, environmental, and safety certifications

6. Management Team

6.1 Organizational Structure

Board of Directors
|
Chief Executive Officer (CEO)
|

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Chief Technical Officer (CTO)	Chief Financial Officer (CFO)	Chief Operations Officer (COO)
Geothermal Team	Finance Team	Operations Team
Drilling/Reservoir	Accounting/Reporting	Plant Management
Power Plant	Treasury/Financing	Maintenance
Environmental	Investor Relations	Health & Safety
Community Relations		Grid Integration

6.2 Key Personnel Requirements

- **CEO:** Energy sector experience, preferably with geothermal or power development background
- **CTO:** Geothermal reservoir engineering, drilling, or power plant experience (Kenyan expertise preferred initially)
- **CFO:** Project finance, infrastructure funding, or energy sector finance experience
- **COO:** Power plant operations, preferably geothermal or similar baseload experience
- **Geothermal Specialists:** Reservoir engineers, geologists, drilling engineers
- **Technical Team:** Mechanical, electrical, and control engineers
- **Local Team:** Rwandan professionals for community relations, permitting, and local content

6.3 Advisory Board

- **Technical Advisors:** KenGen, GDC, or other geothermal experts
- **Financial Advisors:** Infrastructure finance specialists
- **Legal Advisors:** Energy law, PPP specialists, local counsel
- **Environmental Advisors:** ESIA experts, sustainability specialists

6.4 Human Resources Strategy

- **Knowledge Transfer:** Formal programs with Kenyan partners
- **Local Capacity Building:** Training programs, university partnerships
- **Performance Management:** Clear KPIs, regular reviews, competitive compensation
- **Succession Planning:** Identification and development of future leaders
- **Culture:** Safety-first, collaborative, innovative, results-oriented

7. Financial Plan

7.1 Capital Requirements

Phase 1: Foundation and Validation (Q4 2026-Q2 2027)

- **Technical Studies:** \$500,000-\$1,000,000

- Surface geoscience, geochemistry, preliminary modeling
- **Partnership Development:** \$200,000-\$400,000
 - Technical exchange programs, MoUs with Kenyan partners
- **Capacity Building:** \$300,000-\$500,000
 - Training, workshops, knowledge transfer
- **Pilot Design:** \$400,000-\$600,000
 - Direct-use system design for greenhouse/aquaculture
- **Total Phase 1:** \$1.4-\$2.5 million

Phase 2: Pilot Implementation and Exploration (Q3 2027-Q4 2028)

- **Direct-Use Pilot:** \$250,000-\$750,000
 - 100-500kW thermal greenhouse heating system
- **Exploratory Drilling:** \$3,000,000-\$6,000,000
 - 2-3 slim holes + 1-2 production wells at Kinigi/Gisenyi
- **Reservoir Monitoring:** \$500,000-\$800,000
 - Downhole equipment, surface monitoring systems
- **Business Model Development:** \$200,000-\$400,000
 - Scale-up planning based on pilot results
- **Total Phase 2:** \$4.15-\$7.95 million

Phase 3: Commercial Scale (2029 and beyond)

- **5 MW Binary Cycle Plant:** \$12.5-\$20.0 million
 - EPC, grid connection, contingency
- **10 MW Binary Cycle Plant:** \$25.0-\$40.0 million
 - EPC, grid connection, contingency
- **Total Phase 3 (5-10 MW):** \$12.5-\$40.0 million

7.2 Revenue Projections

Assumptions for 5-10 MW Binary Cycle Plant

- **Capacity Factor:** 85% (conservative for baseload geothermal)
- **Degradation:** 0.5% per year
- **Electricity Tariff:** \$0.16/kWh (Rwanda commercial/industrial rate)
- **Plant Lifetime:** 25 years
- **Availability:** 90-95% (including planned maintenance)

Annual Generation and Revenue

Plant Size	Gross Annual Generation	Annual Revenue (Year 1)	Notes
5 MW	37.2 GWh	\$5.95 million	85% CF, \$0.16/kWh

Plant Size	Gross Annual Generation	Annual Revenue (Year 1)	Notes
10 MW	74.4 GWh	\$11.9 million	85% CF, \$0.16/kWh

Note: Revenue assumes full PPA contract. Early years may have lower output during ramp-up.

7.3 Operating Expenses

Annual O&M Costs

- **Fixed O&M:** 1-3% of CAPEX annually
- **Variable O&M:** Minimal (no fuel costs)
- **Parasitic Load:** 10-15% of gross generation (used internally)
- **Insurance:** Standard power plant coverage
- **Property Taxes:** As per Rwandan regulations
- **Administrative:** Management, reporting, compliance

Example O&M for 5 MW Plant (\$16.25M CAPEX)

- **Fixed O&M:** \$162,500-\$487,500/year (1-3%)
- **Variable/Other:** Estimated \$100,000-\$200,000/year
- **Total Annual O&M:** \$262,500-\$687,500/year

7.4 Profitability Analysis

5 MW Plant Example (\$16.25M CAPEX)

Metric	Calculation	Value
Annual Revenue	37.2 GWh $\times$ \$0.16/kWh	\$5.95 million
Annual O&M	2% of CAPEX	\$0.325 million
EBITDA	Revenue - O&M	\$5.625 million
EBITDA Margin	EBITDA/Revenue	94.5%
Simple Payback	CAPEX/EBITDA	2.9 years
25-Year NPV (8%)	DCF of cash flows	~\$55 million

10 MW Plant Example (\$32.5M CAPEX)

Metric	Calculation	Value
Annual Revenue	74.4 GWh $\times$ \$0.16/kWh	\$11.9 million
Annual O&M	2% of CAPEX	\$0.65 million
EBITDA	Revenue - O&M	\$11.25 million

Metric	Calculation	Value
EBITDA Margin	EBITDA/Revenue	94.5%
Simple Payback	CAPEX/EBITDA	2.9 years
25-Year NPV (8%)	DCF of cash flows	~\$110 million

Note: These estimates exclude exploration/drilling risk costs, which are addressed in Phase 1-2.

7.5 Funding Sources

Equity

- **Founders' Capital:** Initial seed funding
- **Strategic Investors:** Energy companies, infrastructure funds
- **Development Finance Institutions:** DFIs taking equity stakes
- **Local Investors:** Rwandan high-net-worth individuals, institutions
- **Employee Ownership:** Potential ESOP for key personnel

Debt

- **Concessional Loans:** World Bank, AfDB, KfW, etc. for exploration and early stages
- **Commercial Debt:** Local and international banks for construction phase
- **Green Bonds:** Climate-focused instruments for environmentally beneficial projects
- **Vendor Financing:** EPC contractor deferred payments

Grants and Technical Assistance

- **Climate Finance:** Green Climate Fund, Clean Technology Fund
- **Technical Assistance:** UNDP, UNIDO, GIZ for capacity building
- **Research Grants:** Academic and international organization funding
- **Risk Guarantees:** Partial risk guarantees from DFIs for exploration drilling

7.6 Financial Sensitivities

Key Variables Impacting Project Economics

1. **Resource Temperature/Flow:** Directly impacts plant size and efficiency
2. **Drilling Success Rate:** Affects capital efficiency and timeline
3. **Electricity Tariff:** Primary revenue driver
4. **Capital Costs:** EPC pricing, contingencies, currency fluctuations
5. **Capacity Factor:** Plant availability and grid constraints
6. **O&M Costs:** Operational efficiency and maintenance effectiveness
7. **Exchange Rates:** For imported equipment and USD-denominated debt

8. **Tax Incentives:** Potential tax holidays or reduced rates for renewable energy

8. Implementation Plan

8.1 Phased Approach

Phase 1: Foundation Building (Q4 2026-Q2 2027) **Objective:** Validate resource potential, establish partnerships, prepare for drilling **Duration:** 6 months **Key Activities:** - Complete geothermal direct-use profitability analysis - Conduct detailed temperature and flow rate measurements at Gisenyi and Bugarama - Engage with agricultural cooperatives and fish farmers near prospect zones - Initiate Kenya-Rwanda geothermal technical exchange program with KenGen and GDC - Begin capacity building for Rwandan professionals through Kenyan training initiatives - Design pilot direct-use system (50-200kW thermal) for greenhouse heating or aquaculture - Finalize solar ROI models for various scales (100kW-10MW) targeting Kigali City - Identify potential sites for pilot solar installations - Engage with MININFRA and REG on net metering and interconnection policies - Develop financing concepts for solar PPAs and green bonds - Conduct preliminary assessment of Kigali's water pumping energy requirements - Identify key pumping stations for energy audits - Research VFDs, high-efficiency pumps, and solar PV integration options

Deliverables: - Validated geothermal resource data - Established technical partnerships - Capacity building program underway - Pilot direct-use system design completed - Financial models for geothermal, solar, and water pumping efficiency

Phase 2: Pilot Implementation (Q3 2027-Q4 2028) **Objective:** Demonstrate viability through pilots, commence exploration drilling **Duration:** 6 months (Note: Timeline appears compressed in source doc - likely 18 months) **Key Activities:** - Launch 100kW thermal greenhouse heating pilot at Gisenyi - Install monitoring systems to track performance and ROI - Develop business model for scale-up based on pilot results - Commence exploratory drilling at Kinigi/Gisenyi using Kenyan best practices and technical partners - Implement reservoir management monitoring from initial wells - Explore aquaculture facility heating pilots alongside greenhouse projects - Launch micro-turbine pilot projects (1-50 kW) for remote communities and industrial applications - Deploy 500kW solar PV system on government building - Implement net metering arrangement with REG - Begin commercial/industrial rooftop solar outreach program - Develop standardized solar PPA templates for large consumers - Conduct grid impact studies for utility-scale solar integration - Conduct water-pumping energy audit at Karege treatment plant - Implement VFDs on major pumping stations - Pilot solar PV integration for daytime pumping operations - Conduct leak detection and repair initiative in distribution network - Explore geothermal-assisted heat pumps for water facilities

Deliverables: - Operational direct-use pilot with verified ROI - Micro-turbine pilot projects operational with performance data - Commenced exploratory drilling program - Reservoir monitoring system operational - Solar PV system generating electricity - Water pumping efficiency improvements demonstrated - Financing mechanisms identified for scale-up

Phase 3: Scale and Replicate (2029 and Beyond) **Objective:** Commercial geothermal power generation and nationwide expansion **Duration:** Ongoing **Key Activities:** - Expand successful pilots to additional sites (Bugarama, Kinigi, etc.) - Develop 1-5MW geothermal binary cycle plants at confirmed resources - Create franchise model for geothermal direct-use systems nationwide - Establish joint ventures with Kenyan partners for larger geothermal power projects (>5MW) - Develop local drilling capacity through partnerships with Kenyan contractors - Implement advanced reservoir management for steam field optimization - Scale to 1-5MW commercial/industrial solar installations - Develop utility-scale (5MW+) grid-connected solar projects with storage - Establish local solar panel assembly and maintenance capabilities - Create solar financing vehicle for residential and small business adoption - Integrate solar with agricultural processing and cold chain facilities - Implement system-wide pump and motor efficiency upgrades - Expand solar PV pumping to additional treatment plants and booster stations - Implement comprehensive pressure management and leak detection programs - Explore hydropower recovery from pressure reduction valves - Develop integrated renewable energy pumping systems (solar + grid + storage)

Deliverables: - Commercial geothermal power plants operational - Nationwide direct-use franchise model - Local drilling capacity established - Integrated renewable energy solutions deployed - Measurable impact on energy access and renewable targets

8.2 Critical Path Items

1. **Resource Confirmation:** Successful exploration drilling with measurable flow/temperature
2. **Partnership Agreements:** Formal technical and commercial agreements with Kenyan partners
3. **Financing Closure:** Commitments from equity and debt providers
4. **Regulatory Approvals:** Environmental, construction, and generation licenses
5. **PPA Execution:** Signed power purchase agreement with REG
6. **EPC Contracting:** Fixed-price turnkey contract for power plant
7. **Grid Connection:** Approved interconnection study and transmission upgrades if needed
8. **Construction Completion:** On-time, on-budget power plant construction
9. **Commissioning:** Successful performance testing and handover
10. **Commercial Operation:** Sustained generation meeting PPA require-

ments

8.3 Milestone Schedule

Milestone	Target Date	Description
Phase 1 Completion	Q2 2027	Foundation building completed, partnerships established
Exploratory Drilling Start	Q3 2027	Commence drilling at Kinigi/Gisenyi
Resource Confirmation	Q2 2028	Measured flow and temperature confirmed
PPA Signed	Q3 2028	Power Purchase Agreement with REG executed
Financial Close	Q4 2028	Debt and equity financing secured
EPC Contract Award	Q1 2029	Engineering, Procurement, Construction contract awarded
Construction Start	Q2 2029	Groundbreaking for geothermal power plant
Mechanical Completion	Q4 2029	Plant construction completed
Commissioning Start	Q1 2030	Systems testing and initialization
Commercial Operation Date (COD)	Q2 2030	Full commercial operation under PPA
Phase 2 Completion	Q4 2028	Pilot implementation completed
Phase 3 Milestone 1	2031	First 1-5MW geothermal plant commissioned
Phase 3 Milestone 2	2033	Joint ventures established with Kenyan partners
Phase 3 Milestone 3	2035	Local drilling capacity operational

9. Risk Management

The detailed risk register is maintained in `RISK_REGISTER.md`. Key risks and mitigations include:

9.1 Critical Risks (Score 20-25)

R1: No Commercially Viable Resource Found

- **Likelihood:** 5 (Near certain >80%)
- **Impact:** 5 (Threatens venture existence)
- **Score:** 25 Critical
- **Mitigation:** Phase 1 structured as falsification test with pre-registered numeric kill criteria; Phase 3 capital contingent on gate; do not commit generation capital before measured flow and temperature exist
- **Residual:** 20 Critical (still high due to inherent exploration risk)

R2: Duplication with EDCL Efforts

- **Likelihood:** 4 (Likely 60-80%)
- **Impact:** 5 (Major — delays or re-scopes a phase)
- **Score:** 20 Critical
- **Mitigation:** Position explicitly as commercialisation layer, not explorer; take proven resource under resource-access agreement; confirm relationship in writing before approaching government
- **Residual:** 8 Medium

9.2 High Risks (Score 12-16)

R3: GRMF Ineligibility

- **Likelihood:** 4 (Likely)
- **Impact:** 4 (Major)
- **Score:** 16 High
- **Mitigation:** Recruit technical partner with East African geothermal delivery record; apply as partner to EDCL's programme
- **Residual:** 8 Medium

R4: No Direct-Use Customer

- **Likelihood:** 4 (Likely)
- **Impact:** 4 (Major)
- **Score:** 16 High
- **Mitigation:** Audit prospective customers' current fuel use before pilot design; reframe as grant-funded demonstration if needed
- **Residual:** 12 High

R5: Unknown Generation Tariff

- **Likelihood:** 4 (Likely)
- **Impact:** 4 (Major)
- **Score:** 16 High

- **Mitigation:** Treat tariff as model output (breakeven) until RURA REFIT level or tender result confirmed
- **Residual:** 8 Medium

R6: Drilling Costs Exceed Budget

- **Likelihood:** 4 (Likely)
- **Impact:** 3 (Moderate)
- **Score:** 12 High
- **Mitigation:** Benchmark from Kenyan actuals with explicit contingency; fixed-price drilling contracts; partial risk guarantee cover
- **Residual:** 6 Medium

R7: Currency Movement

- **Likelihood:** 3 (Possible)
- **Impact:** 4 (Major)
- **Score:** 12 High
- **Mitigation:** Establish PPA currency at term-sheet; local-currency financing for RWF-denominated costs; state FX assumption explicitly
- **Residual:** 6 Medium

R8: Land Access Constraints

- **Likelihood:** 3 (Possible)
- **Impact:** 4 (Major)
- **Score:** 12 High
- **Mitigation:** Confirm tenure and protected-area status early; engage REMA; note northern prospects near Volcanoes National Park
- **Residual:** 6 Medium

R9: Permitting Delays

- **Likelihood:** 4 (Likely)
- **Impact:** 3 (Moderate)
- **Score:** 12 High
- **Mitigation:** Early engagement with REMA and local authorities; sequence licence application ahead of construction decision
- **Residual:** 6 Medium

9.3 Risk Management Approach

Risk Monitoring

- **Quarterly Reviews:** Board-level review of R2-R5, R7 with updates
- **Phase Gates:** Re-rate R1 and R6 based on measured results at Phase 1 completion

- **Annual Full Review:** Complete re-rate aligned with Rwanda’s national planning cycle
- **Ad-hoc Reviews:** On material policy, financing, or technical changes

Risk Mitigation Principles

1. **Exploration Risk:** Start with direct-use applications requiring lower flow/temperature certainty; use government-sponsored drilling for initial confirmation; implement phased investment tied to resource verification
2. **Market Risk:** Secure off-take agreements before major investments; start with applications serving existing markets (food, fish); develop cooperative models with end-users
3. **Technical Risk:** Use proven, off-the-shelf equipment for direct-use applications; engage experienced geothermal consultants for design; implement monitoring and maintenance protocols
4. **Financial Risk:** Seek concessional financing for exploration and pilot phases; structure projects to generate early cash flow from direct use; utilize public-private partnership (PPP) frameworks

10. Economic and Social Impact

10.1 National Energy Contribution

Capacity Contribution

- **5 MW Plant:** ~1% of Rwanda’s 2024 target capacity (512 MW)
- **10 MW Plant:** ~2% of Rwanda’s 2024 target capacity
- **Multiple Plants:** Potential for 10-20% of target capacity with successful exploration

Generation Characteristics

- **Baseload Power:** 90-95% availability complements variable hydro and solar
- **Peak Load Support:** Can provide dispatchable power during peak periods
- **Grid Stability:** Frequency regulation and spinning reserve capabilities
- **Reduced Diesel Displacement:** Direct reduction in expensive diesel generator usage

Energy Security Benefits

- **Import Reduction:** Decreased reliance on imported electricity and diesel fuel
- **Foreign Exchange Savings:** Lower fuel import bills
- **Price Stability:** Insulated from international fuel price volatility
- **Domestic Resource:** Utilizes indigenous energy source

10.2 Employment and Skills Development

Direct Employment

- **Construction Phase:** 50-150 jobs (EPC contractors and subcontractors)
- **Operations Phase:** 3-5 direct jobs per MW (shift-based operations, maintenance, supervision)
- **Indirect Employment:** 2-4 indirect jobs per direct job (suppliers, services, induced)

Skills Development

- **Technical Training:** Geothermal reservoir engineering, drilling, power plant operations
- **Local Content Development:** Partnerships with University of Rwanda, IPRC for curriculum development
- **Knowledge Transfer:** Formal programs with Kenyan experts (KenGen, GDC)
- **Certification Programs:** Operator and technician certification pathways
- **Entrepreneurship:** Spin-off opportunities in geothermal services, maintenance, consulting

10.3 Environmental Benefits

Emissions Reduction

- **CO2 Avoidance:** ~4,200 tons CO2/year avoided per 1MW plant (vs. diesel)
- **Other Pollutants:** Near-zero SOx, NOx, particulates, mercury compared to fossil fuels
- **Lifecycle Emissions:** Minimal compared to fossil fuel alternatives

Resource Stewardship

- **Reinjection:** Sustainable reservoir management through fluid return
- **Surface Footprint:** ~0.5-1 acre per MW (much less than solar or hydro)
- **Water Use:** Minimal consumptive use (binary cycle is closed-loop)
- **Visual Impact:** Low-profile facilities compatible with tourism areas
- **Seismic Monitoring:** Induced seismicity monitoring and management protocols

10.4 Alignment with National Goals

Vision 2020 and NST1

- **Economic Transformation:** Supports industrialization through reliable power
- **Infrastructure Development:** Contributes to energy infrastructure goals

- **Private Sector Engagement:** PPP model encourages private investment
- **Regional Integration:** Supports East African Power Pool objectives

Climate Commitments

- **NDC Contribution:** Direct contribution to 38% emissions reduction pledge
- **Clean Energy Transition:** Advances renewable energy share in generation mix
- **Sustainable Development:** Supports SDG 7 (Affordable and Clean Energy)

Social Development Goals

- **Energy Access:** Enables expansion of grid to underserved areas
- **Job Creation:** Direct and indirect employment opportunities
- **Skills Development:** Technical capacity building for youth and professionals
- **Local Content:** Maximizes Rwandan participation in value chain

11. Conclusions and Recommendations

11.1 Investment Thesis

The geothermal power generation opportunity in Rwanda presents a compelling investment case based on:

1. **Strong National Need:** Clear electricity supply gap and access targets
2. **Significant Resource Potential:** Conservative estimates of 100MW+, with higher estimates suggesting 700MW+
3. **Appropriate Technology:** Binary cycle well-suited to 70-75°C+ resource temperatures
4. **Proven Economic Model:** Attractive returns with 2.7-year payback and 25-year NPV of \$55-\$150 million for 5-10MW plants
5. **Strategic Partnerships:** Opportunity to leverage Kenya's extensive geothermal experience
6. **National Priority:** Alignment with Vision 2020, NST1, and climate commitments
7. **Risk Mitigation:** Phased approach that validates resource before major capital commitment

11.2 Recommended Next Steps

Immediate Actions (0-3 months)

1. **Finalize Team:** Recruit key leadership (CEO, CTO with geothermal experience)

2. **Stakeholder Engagement:** Formal introductions to MININFRA, REG, REMA, RDB
3. **Partnership Outreach:** Initiate discussions with KenGen and GDC for technical cooperation
4. **Preliminary Budget:** Refine Phase 1 budget and identify potential funding sources
5. **Work Planning:** Develop detailed work plan for Phase 1 foundation activities

Short-Term Actions (3-6 months)

1. **Legal Establishment:** Register company, establish governance structure
2. **Initial Technical Work:** Commission desktop geothermal study of prospect zones
3. **Partnership Agreements:** Sign MoUs with Kenyan technical partners
4. **Capacity Building Plan:** Develop training program with Kenyan institutions
5. **Financial Planning:** Prepare preliminary financial model and funding strategy

Medium-Term Actions (6-12 months)

1. **Field Campaign:** Conduct temperature and flow measurements at Gisenyi and Bugarama
2. **Stakeholder Workshops:** Engage local communities and prospective direct-use customers
3. **Drilling Planning:** Prepare exploratory drilling program based on surface results
4. **Funding Applications:** Submit proposals to concessional finance sources
5. **Regulatory Engagement:** Begin preliminary discussions with RURA and REMA on requirements

11.3 Final Statement

Rwanda’s geothermal resources offer a strategic opportunity to develop clean, reliable baseload power that can significantly advance the nation’s energy security, economic development, and climate goals. While exploration risk remains a key challenge, the phased approach outlined in this business plan—beginning with direct-use applications and progressing to power generation as the resource is better characterized—provides a prudent path forward.

By leveraging Kenya’s extensive geothermal experience, implementing proven best practices, and maintaining a disciplined, milestone-driven approach, Rwanda can successfully develop its geothermal potential and establish a sustainable source of clean energy for decades to come. The Shyuha project is well-positioned to lead this effort, creating value for investors while delivering tangible benefits to the Rwandan people and economy.

*This business plan synthesizes research from SUMMARY.md, HOT_SPRINGS_ENERGY_PROFITABILITY.m
PROJECT_TIMELINE.md, RISK_REGISTER.md, and other documents in
the Shyuha repository. Financial estimates are based on available public data
and industry benchmarks. Site-specific feasibility studies are recommended for
accurate project economics.*

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